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Experimental and numerical study on flow characteristic and thermal performance of macro-capsules phase change material with biomimetic oval structure

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ABSTRACT

The emergence of bionics provides new ideas for the innovation of engineering technology, which has been widely used in energy storage, heat transfer enhancement, and solar thermochemical reactions. To improve the flow characteristic and thermal performance of phase change material (PCM) capsules, the idea of heat storage unit with biomimetic oval structure is proposed. An experimental system is developed, and PCM capsules with different structures (bionic-oval, sphere, and ellipse) are prepared by 3D printing. The experimental and numerical analyses show that the unconstrained melting time of oval-shaped capsule is 12% shorter than that of sphere capsule, and the average Nusselt number of oval-shaped capsule is 20% higher than that of sphere capsule. The optimization results show that the oval-shaped capsule can provide a lower resistance coefficient when the long-short ratio and long-symmetry ratio are 1.25 and 1.67, respectively. The research results can provide guidance and new ideas for the application of bionics in the field of energy storage.

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1. Introduction

With the rapid development of modern society, energy shortages and environmental pollution have become the two major problems that need more attention [1–3]. Therefore, it is imperative to develop renewable energy such as solar energy, geothermal energy, and wind energy to optimize the energy composition [4–6]. As one of the most promising renewable energy sources, solar energy has been widely utilized through photoelectricity [7–9], photothermal [10–12], and photochemistry [13–15]. Most recently, the thermal energy storage (TES) system has received considerable attention concerned because of the instability and intermittency of solar energy utilization. The TES system can be divided into three types: sensible heat thermal energy storage (SHTES) [16–18], latent heat thermal energy storage (LHTES) [19–21], and thermochemical heat thermal energy storage (THTES) [22–24].

Nowadays, the LHTES system using phase change materials (PCM) has a large energy storage density than the SHTES system and is more mature than the THTES, which has great potential for practical application [25,26]. The low thermal conductivity of PCM limits their wide application in important engineering [27,28]. Therefore, thermal performance enhancement technology is needed to overcome the disadvantage of low heat transfer efficiency of PCM. These enhancement technologies can be divided into two categories: enhancement of heat transfer area [29,30] and enhancement of thermal conductivity [31,32]. In various LHTES, the packed-bed LHTES has been attracting immense attention for its simple structure and efficient heat transfer [33]. For the packed-bed LHTES, PCM is filled in a large number of macroscopic capsules, which are arranged in storage tanks. Therefore, the heat transfer flow (HTF) flows from the inlet and exchanges heat with PCM capsules, which promotes PCM to complete phase transition, and finally flows out from the discharge outlet. When the capsule size is larger than 1 mm, the shape of the large capsule may be different. The macro-encapsulation PCM is an important subject to improve the efficiency of TES. To date, the most widely used and studied capsule shape is spherical and rectangular due to its simplicity of

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Nomenclatures

c	specific heat, J kg^{-1} K^{-1}
C_D	drag coefficient
D	diameter of the capsule, m
F_D	drag force, N
g	gravity, m s^{-2}
H	total enthalpy
k	thermal conductivity, W m^{-1} K^{-1}
L	latent heat, kJ kg^{-1}
Nu	Nusselt number
p	pressure, Pa
Q	energy transfer from the capsule
Re	Reynolds number
S	momentum source term, N m^{-3}
t	time, s
T	temperature, K
u_i	velocity in i direction, m s^{-1}
U_0	freestream velocity
V	volume, m^3
x, y, z	cartesian coordinates

Greek letters

β	thermal expansion coefficient, K^{-1}
f	liquid fraction
μ	dynamic viscosity, Pa s
θ	location position angle, °
ρ	density, kg m^{-3}

Subscripts

f	heat transfer fluid
lat	latent heat
p	PCM
sen	sensible heat
w	wall

Abbreviations

HTF	heat transfer flow
PCM	phase change material
TES	thermal energy storage

manufacture.

Many researchers have carried out experimental and numerical investigations on the macro-encapsulated PCM to analyze the effects of various operating conditions and geometrical parameters on its thermal performance. Patel et al. [34] developed a visual experimental device to compare the thermal behavior of differently shaped capsules (spherical, cubical, triangular, and plus-shaped) during melting and solidification processes. The results showed that the triangular capsule has better thermal performance, and the total melting and solidification time was 26.8% and 15.8% shorter than the traditional spherical structure, respectively. Erdemir [35] numerically investigated seven geometric modified spherical capsules and compared them with ordinary spherical capsules. It was found that the geometrically modified capsules had higher thermal performance than the ordinary spherical capsules. The thermal performance of the capsules with various pin-fin structures was carried out by Jia et al. [36]. The results showed that the cold charging time of six-fin capsule was 50% shorter than that of non-fin capsule. Shin et al. [37] investigated a shuttle-shaped capsule and applied it to the LHTES system and found the phenomenon of upward and downward motion in solid PCM. Liao et al. [38] proposed a new correlation of effective thermal conductivity through linear regression based on the simulation results of the natural convection inclusion model, which can be used to calculate the melting process of PCM in capsules by conduction control model. Gao et al. [39] observed a new unconstrained melting in sphere capsule due to the existence of void in solid PCM and proposed the relationship between the dimensionless total melting time and the liquid fraction.

Besides, the flow characteristics of the capsule also have a certain influence on the energy consumption of the TES system. Scholars have also studied the flow characteristics of different structure capsules. Li et al. [40] numerically studied the flow and heat transfer past two side-by-side spheres using the large eddy simulation (LES) method at Re = 5000. They found that a biased gap flow that occurs when G/D = 1.25 influences the wake zone length and the drag coefficients on the spheres. Zare et al. [41] numerically and experimentally investigated heat transfer of a multi-lobe particle (tri-lobe, tetra-lobe, and penta-lobe) affected by wall in cross and axial-flow. The results depicted that wall effects can be ignored when the particle diameter ratio was greater than 5. Toghraie et al. [42] numerically studied the effects of size and arrangement of particle on the flow and heat transfer in packed-bed. They found that the Nusselt number in cross-flow is larger than that in axial-flow. Wang et al. [43] proposed an equivalent network model based on Voronoi tessellation to predict the flow field. They found that the flow rate of a channel depends on the hydraulic resistances of adjacent channels rather than just on the channel itself. A numerical study of PCM capsules in the forced convection with spherical and red-blood shape was presented by Cheng et al. [44]. They found that there was little difference in the resistance coefficient between the two, and the average Nusselt number on the surface of red blood cell cysts was smaller than that of spherical cysts. Guo et al. [45] investigated the ascending and melting characteristics of spherical ice particles under different superheat and supercooling degrees. The numerical results exhibited that the ice particles were spherical, elliptical, barrel and oval in order during the melting process.

The bionic research provides new options for effectively solving major problems in current economic and social development [46]. Natural evolution has caused long-term interactions between organisms and the environment, and many excellent functions and perfect structures have emerged. At present, there are two kinds of bionics in the field of energy storage: structural bionics and functional bionics. Liang et al. [47] proposed a novel spectral splitting method based on front-located wavelength-sized TiO2 biomimetic moth-eye nanophotonic structure, which can inhibit the increase of charge carrier capture and recombination. Cheng et al. [48] proposed an RBC-shaped capsule and compared the thermal performance of different structures by experiment and simulation. The results indicated that the thermal performance of RBC-shaped capsule was better than sphere capsule. Wang and Shi et al. [49–51] proposed the idea of an ordered hierarchical structure in solar thermal chemical reaction chambers to improve the conversion efficiency of solar-driven methane reforming. The results showed that the effective combination of different pore sizes can significantly improve methane conversion. Given the lack of reliable heat transfer network design criteria, Liu et al. [52] proposed an optimization criterion for simulating the vein generation process, which can prolong the working time of electronic devices. Dong et al. [53] proposed the idea of a bionic leaf hierarchical porous structure LHTES. The results showed that the thermal

performance of the optimized packed-bed LHTES has been enhanced. Tian et al. [54] proposed a bionic topology optimization method considering different objective functions and heat transfer mechanisms to improve heat storage/heat release performance. The results indicated that the optimized fin structure can reduce the melting and solidification time.

The literature review indicates that there are two conventional methods to improve the performance of LHTES: increasing the thermal conductivity of PCM and extending the heat transfer area, few innovations have been made in the structure of PCM capsule. The heat storage unit in the packed-bed has a similar structure to the oval in the ovipara. Considering the unique advantages of the oval structure in the reproductive process and growth process, the idea of biomimetic oval-shaped structure is proposed to improve the flow characteristic and thermal performance of PCM capsule based on the principle of bionics. The influence of packaging forms (bionic-oval, sphere, and ellipse) and geometric parameters of PCM capsules on the flow and heat transfer process are analyzed by experiment and numerical study. The research results can provide guidance and new ideas for the application of bionics in the field of energy storage.

2. Experimental system and procedure

2.1. Formation of the oval-shaped capsule

A typical animal egg has two main advantages: reproductive advantage and growth advantage. The reproductive advantage makes the ovoid structure produce from the ovipara with less flow resistance. The growth advantage makes the yolk absorb nutrition more evenly during the growth process. The heat storage unit in the packed-bed has a similar structure to the oval of oviparous animals. The reproductive process of ovipara is analogous to the flow process of capsules in the TES system, and the hatching process of ovipara is analogous to the endothermic process inside PCM capsules. Therefore, it can be preliminarily considered that the bionic oval-shaped heat storage unit is feasible and reasonable to replace the traditional sphere heat storage unit. An oval is a closed curve in a plane that resembles the outline of an egg, and an ovoid is a surface in 3-dimensional space generated by rotating an oval curve about one of its axes of symmetry. The shape of an egg is approximated by the "long" half of a prolate spheroid, joined to a "short" half of a roughly spherical ellipsoid, or even a slightly oblate spheroid [55]. These are joined at the equator and share a principal axis of rotational symmetry. It can also be used to describe the 2-dimensional figure that, if revolved around its major axis, produces the 3-dimensional surface.

Three parameters are needed to determine an oval curve: the long radius a, the short radius b, and the symmetry radius c. As shown in Fig. 1, the oval becomes an ellipse when $a = b \neq c$, and becomes a circle when $a = b = c$. The oval equation with binary cubic can be written as [56]:

$$\frac{x^2}{(a-b)x+ab}+\frac{y^2}{c^2}=1, x\in[-b,a] \tag{1}$$

or:

$$\begin{cases} x = c\cos t, \\ y = \frac{1}{2}\left[-(a-b)\sin t+\sqrt{(a-b)^2\sin^2 t+4ab}\right]\sin t \end{cases} t\in[0,2\pi] \tag{2}$$

Therefore, the volume and surface area of the three-dimensional ovoid obtained by rotating the plane figure around the x-axis can be calculated by Eqs. (3) and (4).

$$\begin{aligned} V &= \pi\int_{-b}^{a} y^2(x)\mathrm{d}x = \pi c^2\int_{-b}^{a}\left[1-\frac{x^2}{(a-b)x+ab}\right]\mathrm{d}x \\ &= \pi c^2\left[\frac{a+b}{2}+\frac{ab(a+b)}{(a-b)^2}-\frac{2a^2b^2}{(a-b)^3}\ln\frac{a}{b}\right] \end{aligned} \tag{3}$$

$$\begin{aligned} S &= 2\pi\int_{-b}^{a}|y(x)|\sqrt{1+y'^2(x)}dx \\ &= \pi c\int_{-b}^{a}\frac{\sqrt{4\left[-x^2+(a-b)x+ab\right]\left[(a-b)x+ab\right]^3+c^2x^2\left[(a-b)x+2ab\right]^2}}{\left[(a-b)x+ab\right]^2}dx \end{aligned} \tag{4}$$

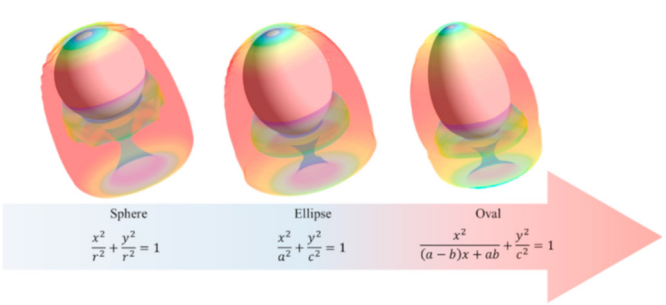


Fig. 1. The evolution of capsule from sphere to oval.

To analyze the influence of different structures on their thermal performance, 3D printing methods are used to prepare different types of PCM capsules (sphere, ellipse, and oval). An additional tube is set on the top of the heat storage unit to fill PCM and thermocouples, and achieve constrained/unconstrained melting. The

Table 1
Parameters of differently shaped capsules.

Name	a (mm)	b (mm)	c (mm)	Surface area (mm^2)
Sphere	34.3	34.3	34.3	14792.8
Oval	50.0	40.0	30.0	15219.0
Ellipse	40.5	40.5	31.6	14904.4

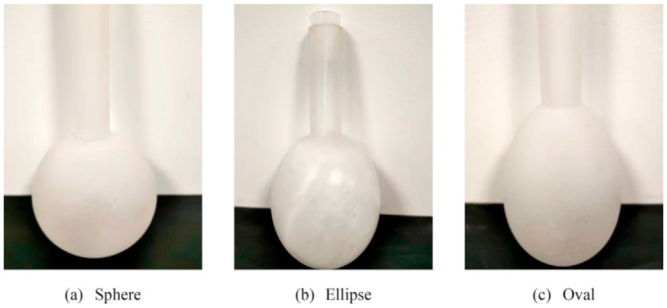


Fig. 2. Photo of the capsules.

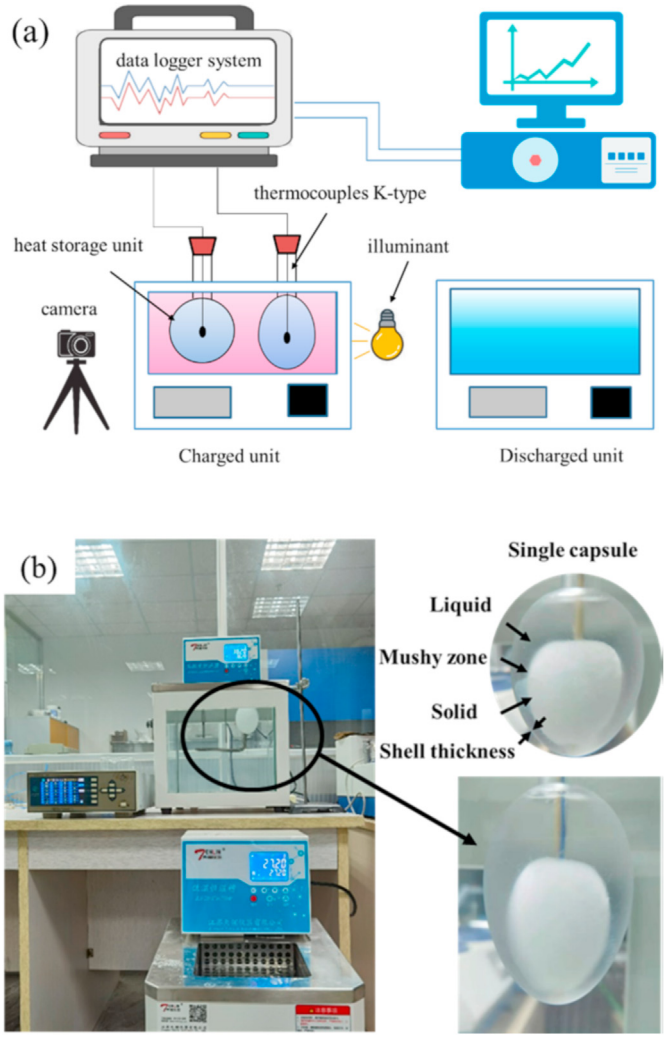


Fig. 3. Schematic diagram (a) and real-picture (b) of the experimental system.

specific geometric parameters excluding long rods are shown in Table 1, and different heat storage units have the same PCM volume. Fig. 2 is a photo of the printed capsules, the geometric parameters *a*, *b*, and *c* of oval structure are setted as 50, 40 and 30 respectively, which is similar to the common oval. To achieve a better visualization effect, the homo-thermal water bath is made of glass, and the PCM capsule is made of translucent photosensitive resin with a thickness of 2 mm.

Table 2
Thermo-physical properties of materials [57].

Properties	*n*-octadecane	resin
Density (kg/m^3)	774.5	1200
Melting temperature (K)	301.35	/
Specific heat (J/kg·K)	2330.0	900
Latent heat (kJ/kg)	243.5	/
Thermal conductivity (W/m·K)	0.1505	0.4
Dynamic viscosity (kg/m·s)	3.86×10^{-3}	/
Coefficient of thermal expansion (K^{-1})	9.42×10^{-4}	/

2.2. Experimental system

The visualization of the liquid-solid interface during the melting and the solidification processes is established. The experimental system used in this paper is shown in Fig. 3. The heat transfer performance test system primarily consists of two homo-thermal water baths (high-temperature and low-temperature), a data acquisition device, a computer, a digital camera, and some test capsules filled with *n*-octadecane. The thermophysical properties of the PCM and capsule are listed in Table 2. Fig. 3 is a real-picture of the experimental system. The visualization tank is made up of glass for better visualization. In the experimental stage, the internal closed environment is realized by a rubber plug. The water line of the homo-thermal water bath is set to just exceed the heat storage unit so that no additional tubes participate in heat exchange. Therefore, it can be considered that the existence of the tube will not affect the experimental results.

The experimental steps are as follows: First, an initial sub-cooling temperature of 1 K for the solidified PCM is maintained of the low-temperature homo-thermal water bath, and the high-temperature homo-thermal water bath is set to a temperature that is 10 K higher than the PCM melting temperature. Second, the capsule filled with PCM is submerged into the low-temperature water bath to achieve a uniform initial temperature. Then, the capsule is quickly moved from the taken from the low-temperature water bath to the high-temperature water bath to start the melting experiment. The capsule is photographed and recorded regularly through the visualization window until it completely melts, and the liquid fraction of capsule is calculated by the software ImageJ developed at the National Institutes of Health according to the difference of color gamut.

As shown in Fig. 4, the experimental process consists of two parts: constrained melting and unconstrained melting. For the constrained melting, the thermocouple is fixed on a rod, which is inserted into the capsule. The rod fixes the thermocouple in the center of the capsule and also fixes the solid PCM onto the rod to

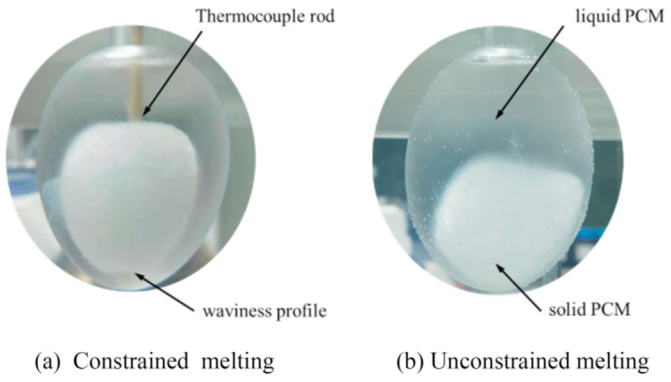


Fig. 4. Melting inside the capsule: (a) constrained melting, (b) unconstrained melting.

prevent the solid PCM from sinks to the bottom of the capsule. For the unconstrained melting, temperature measurement of PCM in the ball is not involved, which means that thermocouple is not used. As shown in Fig. 4(b), the solid PCM sinks into the bottom of the capsule and contacts the shell [57].

3. Mathematical model

3.1. Model description and governing equations

The thermal performance and flow characteristics of PCM capsules with different structures are established and compared. The numerical model is divided into two parts: 1) forced convection of external flows over capsules with different structures and 2) internal melting process of the single PCM capsule with different structures. For the first part, a three-dimensional numerical domain is built outside the heat storage unit, and the size of the numerical domain is 20D × 10D × 10D (x, y, z). For the second part, the thermal performance of oval-shaped capsules with different geometrical parameters is studied, and the melting processes are compared. All oval-shaped capsules have the same volume, and the geometric parameters of oval-shaped capsules will be introduced in Section 4.3.

To simplify the problem, the following assumptions are made in this study:

- The properties of PCM are homogeneous and isotropic.
- The thermophysical properties of the PCM are assumed to be constant throughout the process.
- The boussinesq approximation is applied to consider the effects of natural convection during the phase change process, in which density change is only considered with buoyancy force.
- The HTF is assumed as incompressible in the duct system with temperature-independent properties as the pressure change is very small and the density of HTF is considered unchanged.
- The external flow does not consider gravity and other external forces.

Based on the above assumptions, the continuity, momentum, and energy equations of the single capsule model and the external flow model of the capsule can be written as:

For the HTF [40]:

Continuity equation:

$$\frac{\partial \rho_f}{\partial t}+\frac{\partial}{\partial x_i}\left(\rho_f \overline{u}_i\right)=0 \tag{5}$$

Momentum equations:

$$\frac{\partial}{\partial t}\left(\rho_f \overline{u}_i\right)+\frac{\partial}{\partial x_j}\left(\rho_f \overline{u}_i \overline{u}_j\right)=\frac{\partial}{\partial x_j}\left(\sigma_{ij}\right)-\frac{\partial \overline{p}}{\partial x_i}-\frac{\partial \tau_{ij}}{\partial x_j} \tag{6}$$

$$\sigma_{ij}=\left[\mu\left(\frac{\partial \overline{u}_i}{\partial x_j}+\frac{\partial \overline{u}_j}{\partial x_i}\right)\right]-\frac{2}{3}\mu\frac{\partial \overline{u}_l}{\partial x_l}\delta_{ij} \tag{7}$$

$$\tau_{ij}=\rho_f\overline{u_i u_j}-\rho_f \overline{u}_i \overline{u}_j \tag{8}$$

Energy equation:

$$\frac{\partial\left(\rho_f H\right)}{\partial t}+\frac{\partial\left(\overline{u}_i \rho_f H\right)}{\partial x_i}=k\frac{\partial^2 T}{\partial x_i^2} \tag{9}$$

For the PCM:

Continuity equation:

$$\frac{\partial \rho_p}{\partial t}+\frac{\partial\left(\rho_p \overline{u}_i\right)}{\partial x_i}=0 \tag{10}$$

Momentum equations:

$$\frac{\partial\left(\rho_p \overline{u}_i\right)}{\partial t}+\frac{\partial\left(\rho_p \overline{u}_i \overline{u}_j\right)}{\partial x_j}=\mu\frac{\partial^2 \overline{u}_i}{\partial x_j \partial x_j}-\frac{\partial \overline{P}}{\partial x_i}+\rho g_i \beta\left(T-T_o\right)+S_i \tag{11}$$

Energy equation:

$$\frac{\partial\left(\rho_p H\right)}{\partial t}+\frac{\partial\left(\rho_p \overline{u}_i H\right)}{\partial x_i}=\frac{\partial}{\partial x_i}\left(k\frac{\partial T}{\partial x_i}\right) \tag{12}$$

The enthalpy-porosity method is used to simulate the phase transition process of PCM. According to the enthalpy distribution, only one regional governing equation is needed to determine the phase transition interface. The symbol S is the source term (N/m^3) of porosity in the mushy zone, the enthalpy-porosity technique treats the mushy region (partially solidified region) as a porous medium [58].

$$S_i=\frac{(1-f)^2}{f^3+\varepsilon}\mathrm{A}_{\mathrm{mush}}\left(u_i-u_{i-\mathrm{pull}}\right) \tag{13}$$

In the above equation, the ε is a small number (0.001) which can prevent division by zero. A_{mush} is a parameter of the mushy zone which is equal to 10^6. u_{pull} and v_{pull} are the solid velocities produced by pulling solidified material out of the domain. The total enthalpy of PCM, H, can be calculated as the sum of sensible heat h_{sen} and latent heat h_{lat} [59]:

$$H=h_{\mathrm{sen}}+h_{\mathrm{lat}} \tag{14}$$

$$h_{\mathrm{lat}}=fL \tag{15}$$

where L is the latent heat during the phase change process, the liquid fraction (f) is a number that varies from 0 (solid state) to 1 (fluid state). The liquid fraction can be defined as follows:

$$f=\begin{cases}0 & h<h_s=c_p T_m\\ \dfrac{h-h_s}{L} & h_s<h<h_s+\mathrm{L}\\ 1 & h_s+\mathrm{L}<h\end{cases} \tag{16}$$

For the boundary conditions, the domain outlet is set as the pressure outlet with 0-gauge backflow pressure. The four walls are used the same flow velocity as the inlet, and the thermal boundary conditions are set as adiabatic boundaries. The surfaces of capsule are no-slip, with constant temperature boundaries at 11 K higher than the inlet temperature. The initial conditions can be defined as:

$$t=0,\ T_f=T_p=T_{\mathrm{ini}},\ v=v_{\mathrm{in}} \tag{17}$$

3.2. Methodology

In this study, the governing integral equations for the conservation of mass, momentum, energy, and other scalars are solved through the finite volume method. The velocity-pressure coupling is resolved by adopting the Semi-Implicit Pressure-Linked Equation (SIMPLE) algorithm, and the Pressure Staggering Option Scheme (PRESTO) scheme is taken into account to solve the pressure correction equation. The QUICK (Quadratic Upstream Interpolation for Convection Kinematics) differencing scheme is employed to discretize the energy and momentum equations. The consequences of grid size and time step are examined at the initial stages of the simulation. The careful examination revealed that optimum values for the grid size and time step are 5×10^{-4} m and 0.1 s. The convergence of the numerical solution is examined at each time step, and the convergence criteria are set as 10^{-4} for the continuity equation, and 10^{-6} for the momentum equation and energy equation, respectively.

3.3. Model validation

Based on the above methodology, the developed model of single capsule is validated by the experimental results in this study. The flow and heat transfer process of the capsule under forced convection is verified by other researchers. A series of numerical calculations are carried out at Reynolds number range from 20 to 300 to verify the flow and heat transfer of the numerical model. The predicted mean drag coefficient, C_D, and the Nusselt number, *Nu*, are compared with the experimental data of Whitaker [60] and Cheng [61]. As shown in Fig. 5, the numerical results in this paper agree well with the experimental data, which validate the feasibility of this method to predict the heat transfer and flow characteristics for flows past these capsules.

For the melting process, the initial temperature of the PCM and HTF is maintained at 300.35 K and 311.35 K, respectively. The obtained experimental data are compared with the numerical results, the liquid fraction and temperature of oval-shaped PCM capsule during the melting process with time are presented in Fig. 6. It can be seen from Fig. 6 that the temperature and liquid fraction evolution of this model is in good agreement with the experimental results, which indicates the acceptable agreement between the numerical and experimental results. The developed model, therefore, can be employed for the thermal performance investigation of the differently shaped capsules.

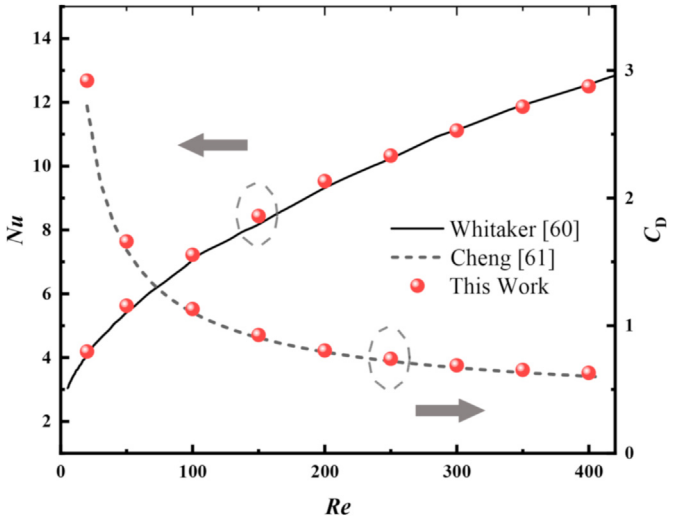


Fig. 5. Comparison of numerical model and literature data in drag coefficient and Nusselt number.

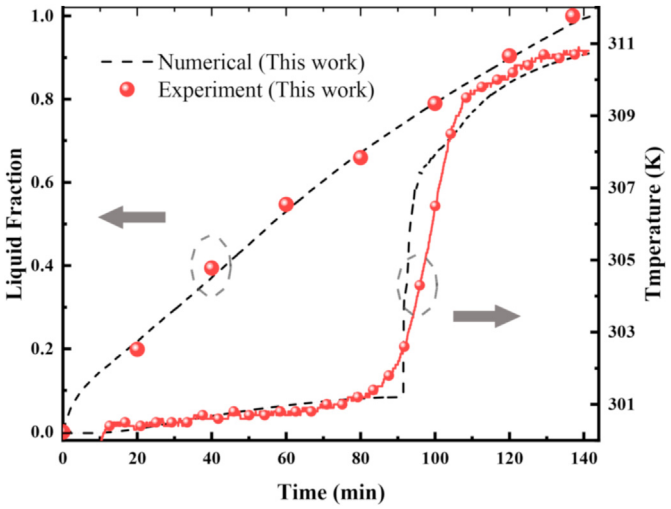


Fig. 6. Comparison of numerical and experimental results in temperature and liquid fraction.

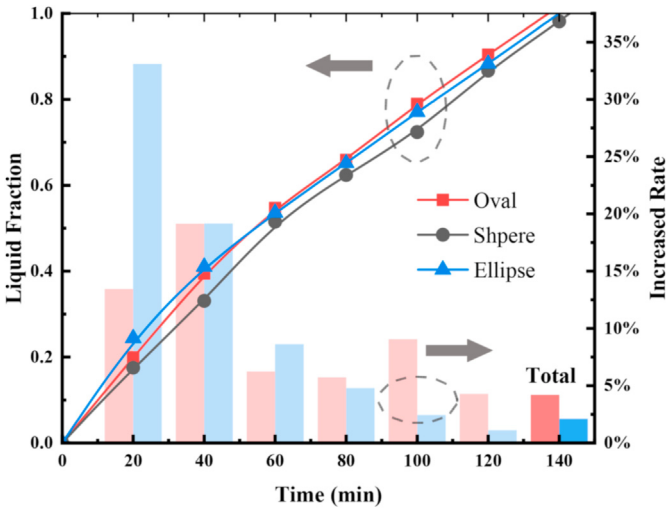


Fig. 7. Liquid fraction versus time in constrained melting.

4. Results and discussion

4.1. Experimental study on melting process of biomimetic oval-shaped capsules

Fig. 7 shows the liquid fraction versus time in constrained melting with different structures (bionic-oval, sphere, and ellipse). It can be found from Fig. 7 that rapid melting was observed in the early stage of melting process. After a while, the melting rate becomes uniform. The oval-shaped capsule has the shortest melting time of 137 min, and the ellipse and sphere capsules have a total melting time of 140 and 143 min, respectively. Fig. 7 also provides the comparison of increased rates of oval-shaped and ellipse-shaped capsules in different stages. Since the oval-shaped and ellipse-shaped capsules have a larger heat transfer area than the sphere capsule under the same volume, there is an increased rate of about 20% in the initial stage where heat conduction is the main

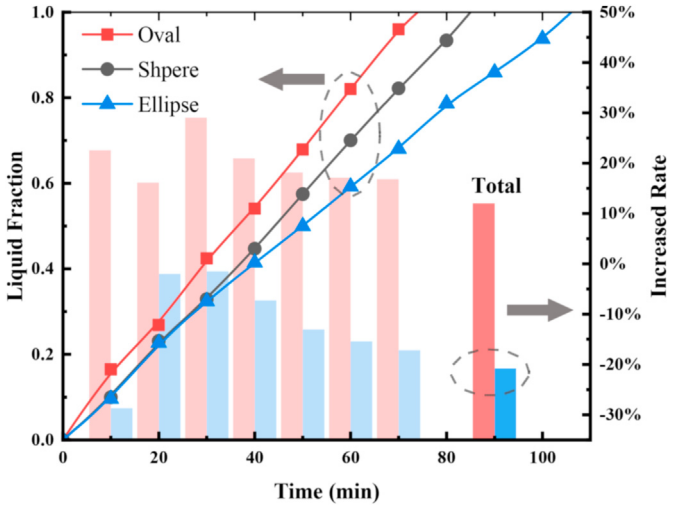


Fig. 8. Liquid fraction versus time in unconstrained melting.

heat transfer mode. When the heat transfer is dominated by natural convection, the increased rate is about 10%, and the complete melting time of oval-shaped and ellipse-shaped capsules increased by 4.2% and 2.2%, respectively.

Fig. 8 indicates the liquid fraction versus time in unconstrained melting with different structures (bionic-oval, sphere, and ellipse). Since the heat conduction process is the main heat transfer mode in unconstrained melting, the liquid fraction varies uniformly throughout the process. The oval-shaped capsule has the shortest melting time of 74 min, and the ellipse and sphere capsules have a total melting time of 106 and 85 min, respectively. It can be seen from Fig. 8 that the oval-shaped capsule can shorten the melting time by 12%. The melting time of the ellipse-shaped capsule is prolonged by 20%, which is not conducive to the melting of the capsule. This is mainly due to the small contact area at the bottom of the ellipse, and the heat conduction process is weaker than the other two structures.

The experimental results show that the melting rate of unconstrained melting in the single capsule seems to be faster than that of the constrained melting. This is due to the high heat transfer rate from translucent photosensitive resin (shell) to solid PCM. The heat transfer schematic diagrams of the two melting processes are shown in Fig. 9. During the initial stage, the heat transfer is carried out by heat conduction, and as the melting process, natural convection becomes the dominant mode of heat transfer. Due to the low thermal conductivity of liquid PCM, the heat transfer process

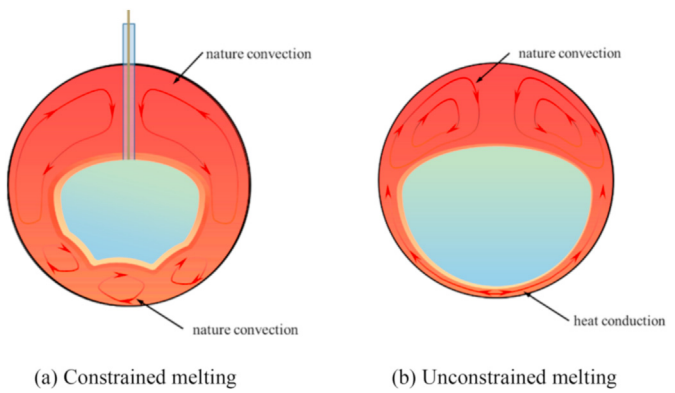


Fig. 9. Schematic diagram of heat transfer in constrained and unconstrained melting.

becomes slow. Natural convection means that the hot liquid PCM rises along the hot wall under the action of temperature difference, while the cooler liquid in the center flows downward to replace the warmer fluid. This creates an unstable fluid cycle in the sphere, which is called buoyancy driven convection. The larger the space at the bottom of the capsle, the stronger the natural convection effect. Under the constraint conditions, the melting front is uniform at the top and wavy at the bottom. For the unconstraint melting, the solid PCM sinks to the bottom of the capsule. The liquid PCM at the bottom is pushed up to the upper half through the side. There is heat conduction between the solid PCM at the bottom and the shell, while there is natural convection in the liquid region of the upper part. Since the heat transfer ability of heat conduction process is greater than that of natural convection, the melting time required for unconstrained melting is less than that of constrained melting. The heat conduction rate increases with the contact area between solid PCM and shell.

4.2. Numerical study on flow characteristic of biomimetic oval-shaped capsules

After the validation of the numerical method, the flow and heat transfer past different PCM capsules are predicted. The main simulation parameters Reynolds number (Re), drag coefficient (C_D), and Nusselt number (Nu) in present study are defined as follows:

$$Re = \frac{\rho U_0 D}{\mu}, C_D = \frac{2F_D}{\rho U_0^2 D}, Nu = \frac{Q}{\pi D(T_w - T_0)k_f} \tag{18}$$

where U_0 is the freestream velocity, F_D is the drag force on the capsule, Q is the energy transfer from the capsule, T_w is the surface temperature of the capsule, T_0 is the temperature of inlet fluid, and k_f is the thermal conductivity of heat transfer fluid.

The external resistance of capsule has an important influence on the flow state, Fig. 10 shows the drag coefficient and drag force on the surface of capsule with different structures at various Reynolds numbers. It can be seen from the figure that when the Reynolds number is at a lower level (20–100), the sphere capsule has a lower drag coefficient, and the drag force of the three structures is the same. With the increase of Reynolds number, the drag coefficient decreases and the drag force increases. When the Reynolds number is greater than 100, the oval-shaped capsule has the lowest drag coefficient and drag force, which can be considered that the oval structure has the best flow characteristics among the three

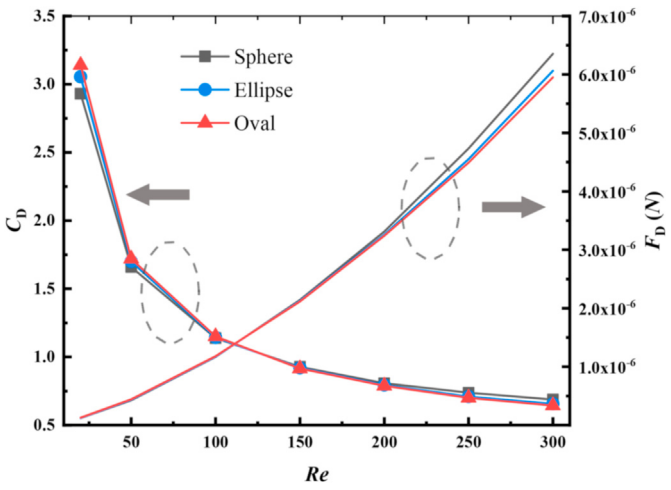


Fig. 10. Mean drag coefficients and drag force of capsule at various Re numbers.

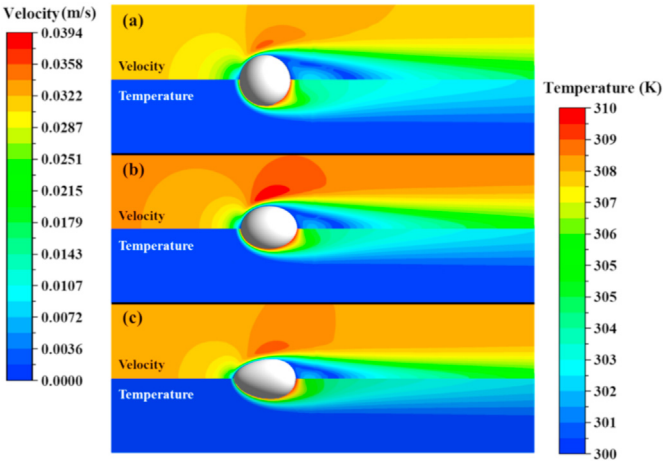


Fig. 11. Velocity and temperature distribution for: (a) sphere, (b) ellipse, and (c) oval at Re = 150.

structures.

Fig. 11 illustrates the velocity and temperature distribution for different structures of capsules at Re = 150. Due to the symmetry of geometry, the velocity and temperature contour are also symmetrical. It can be seen from Fig. 11 that a stagnation zone and a boundary layer are formed at the front contact surface due to the contact between the HTF and the capsule. Along the direction of fluid flow, the velocity boundary layer gradually thins and separates along the wall, forming a vortex zone at the back of the capsule. The existence of a thermal boundary layer can also be observed in the figure, and the thickness of the thermal boundary layer increases gradually along the flow direction. Due to the flow advantages of oval structure, it has the smallest velocity stagnant areas and the smallest width of thermal boundary layer and velocity boundary layer.

Fig. 12 represents the instantaneous 3D streamlines and vortex structures for different structures at Re = 150, wherein the line, translucent region, and color represent streamline, vortex core region, and strength, respectively. It can be observed from Fig. 12 that the oval structure has the smallest windward stagnation zone. In addition, the streamlined structure of the oval-shaped capsule is more aerodynamic. Due to the long front area and short tail region, it has less resistance when impacted by external fluid. Therefore, the flow characteristics of the structure are better than that of the sphere structure and the ellipse structure.

Fig. 13 illustrates the Nusselt number of different structures at various Reynolds numbers. It can be concluded from Fig. 13 that the Nusselt number increases with the Reynolds number, but the increased rate decreases gradually. Fig. 13 also shows that under the same Reynolds number, the oval-shaped capsule has the largest Nusselt number, followed by the ellipse-shaped capsule, and the traditional sphere capsule is the third. The Nusselt number of the oval-shaped capsule is 20% higher than that of the sphere capsule.

Fig. 14 shows the local Nusselt number of the three structures at Re = 150. The location position angle, θ, is the angle between the x-axis of the flow field and the connecting line between the center and surface of the capsule. θ start from the opposite direction of the external fluid. It can be seen from the figure that the windward side has the largest local Nusselt number, which is due to the vertical contact between the shell and HTF. It decreases gradually with the decrease of contact angle between HTF and capsule. When the boundary layer is separated, the local Nusselt number is minimum. Then, due to the existence of tail vortex, the local Nusselt number rises and then decreases near the axis symmetry of the vortex. Due to the symmetry of the three structures, only half of the region is shown in the figure. Similar to the above conclusion, the oval-shaped structure has the best heat transfer performance. The sphere structure has the worst heat transfer performance and only has a little advantage in the tail vortex area.

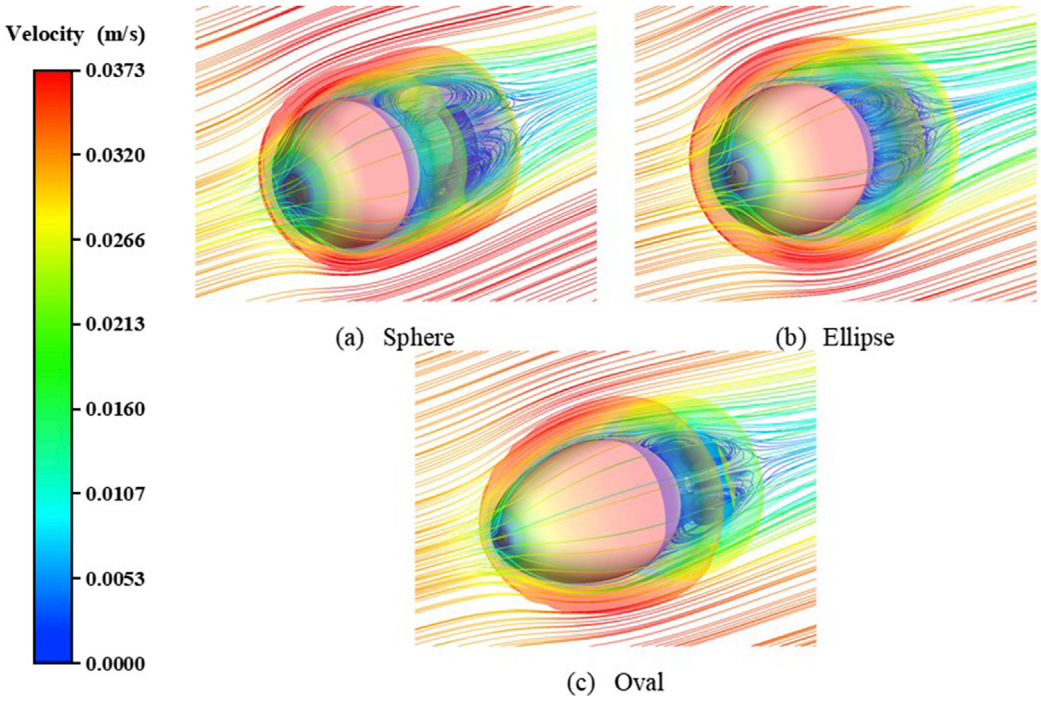


Fig. 12. Instantaneous 3-D streamlines and vortex structures for different structures at Re = 150.

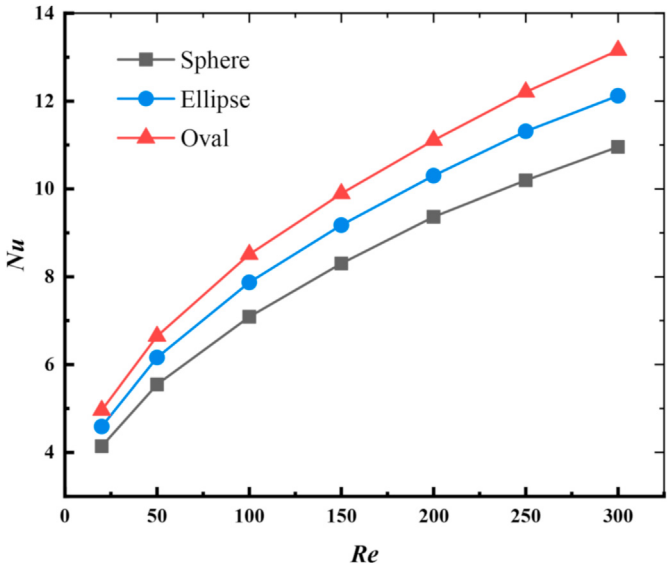


Fig. 13. Nusselt number of different structures at various Reynolds numbers.

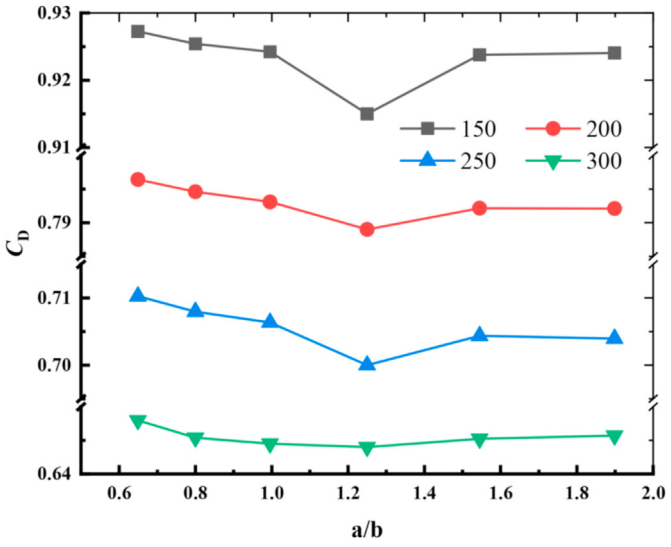


Fig. 15. The effect of a/b on the drag coefficient of oval-shaped capsule. ($150 \leq Re \leq 300$).

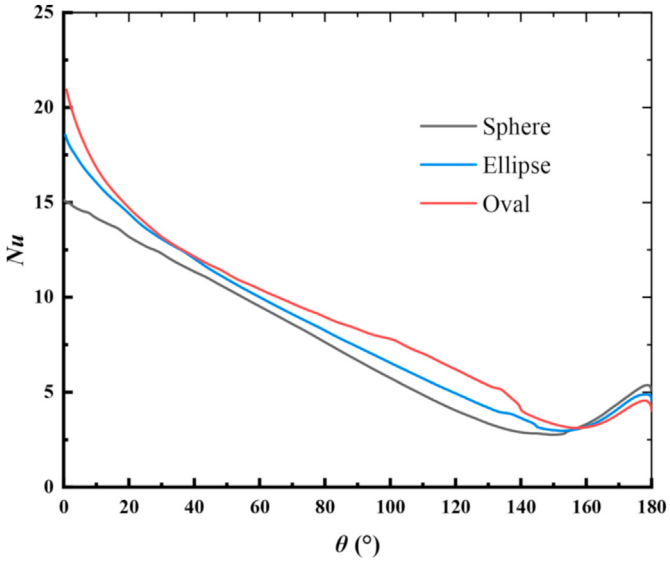


Fig. 14. Local Nusselt number of capsules at different location position angles (Re = 150).

4.3. Structural optimization of biomimetic oval-shaped capsules

The advantages of the biomimetic oval structure in forced convection and heat transfer have been studied through experiments and numerical analyses bove. It can be seen from Eq. (1) that the structure of oval is determined by three geometric parameters. Therefore, it is necessary to analyze the changes in the flow and heat transfer characteristics of the oval structure under different parameters. Under the same volume, the numerical analysis is carried out by the control variable method. Table 3 shows the geometric parameters of the bionic oval-shaped capsule, where *a* and *b* represent the long radius and short radius, *c* represents the symmetric radius.

Fig. 15 shows the effect of a/b on the drag coefficient of the biomimetic oval-shaped capsule under different Reynolds numbers. It can be observed from Fig. 15 that for the same capsule, the drag coefficient decreases with the Reynolds number. For the same Reynolds number, the drag coefficient initially decreases and then increases with the increase of a/b. When the Reynolds number is between 150 and 300, the capsule with the ratio of long-radius to short-radius of 1.25 has the minimum drag coefficient.

Fig. 16 shows the effect of a/b on the Nusselt number of capsules when the Reynolds number is between 20 and 300. It can be seen from Fig. 16 that the Nusselt number increases with the Reynolds number, and the increase rate slows down with the increase of Reynolds number. For the same Reynolds number, the larger of a/b, the larger of the Nusselt number. That is, the larger the windward area is, the larger the Nusselt number is. But the curves are almost identical, which means that a/b has little effect on the Nusselt number of an oval capsule.

To show the effect of geometric parameters on the melting of a single oval-shaped PCM capsule more clearly, the melting time, surface-volume ratio, and average surface heat flux are integrated into one graph [48]. As shown in Fig. 17, when a/b increases from 0.649 to 1.898, the total melting time decreases from 9090 s to 8390 s. The surface-volume ratio decreases first and then increases with the increase of a/b. When the ratio of long-radius to short-radius is 1.25, the capsule has a lower surface-volume ratio and higher average surface heat flux, which are 89.78/mm and 271.56 W/m^2, respectively. It can be concluded that the oval-shaped

Table 3
Geometrical parameters of biomimetic oval-shaped capsules.

Geometry	Ratio	Surface area (cm^2)	Ratio	Surface area (cm^2)	Ratio	Surface area (cm^2)
a/b	1.898	153.69	1.545	152.72	1.250	152.19
	0.996	151.93	0.800	152.19	0.649	152.71
a/c	2.098	156.42	1.877	154.25	1.667	152.19
	1.456	150.76	1.064	148.75	0.880	148.47

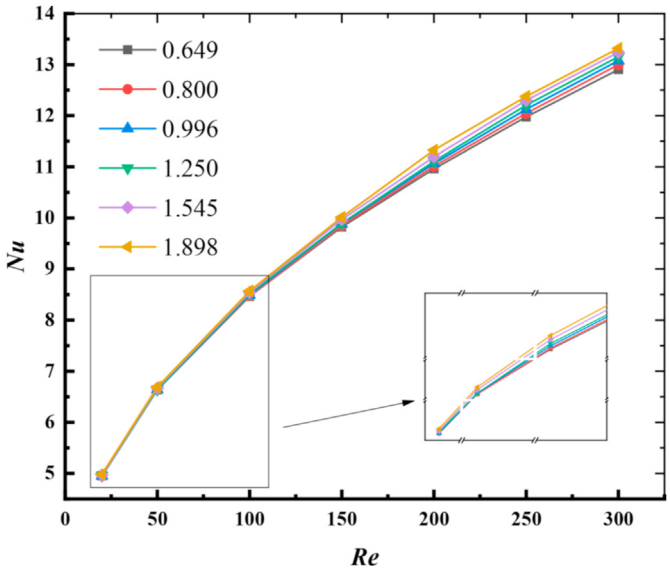


Fig. 16. The effect of a/b on the Nusselt number of oval-shaped capsules. ($20 \leq Re \leq 300$).

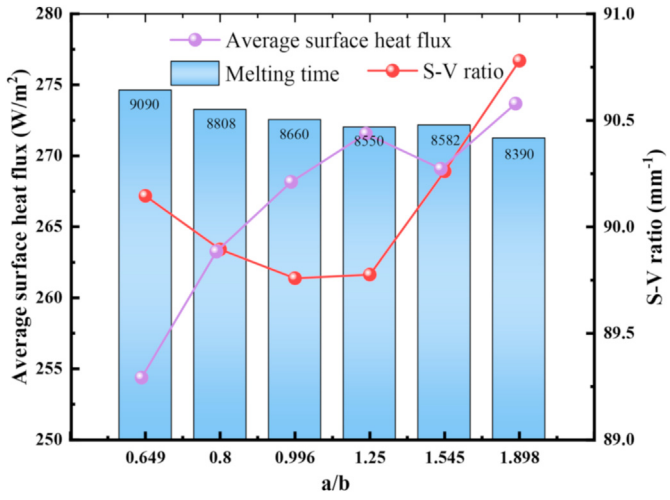


Fig. 17. The effect of a/b on the thermal performance of oval-shaped capsules.

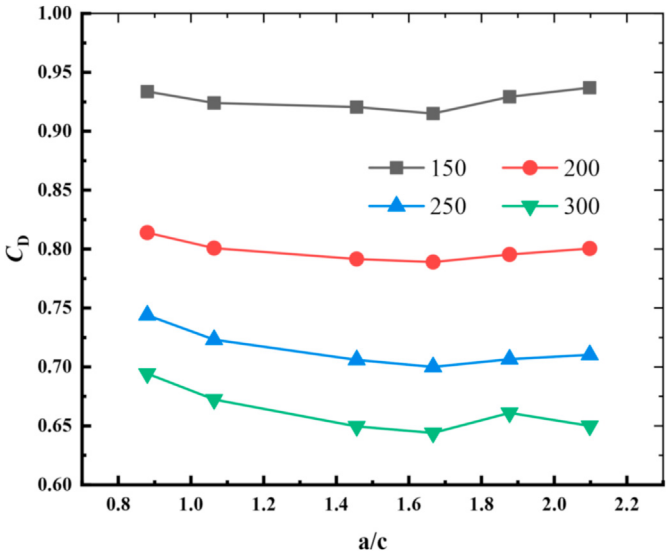


Fig. 18. The effect of a/c on the drag coefficient of oval-shaped capsules. ($150 \leq Re \leq 300$).

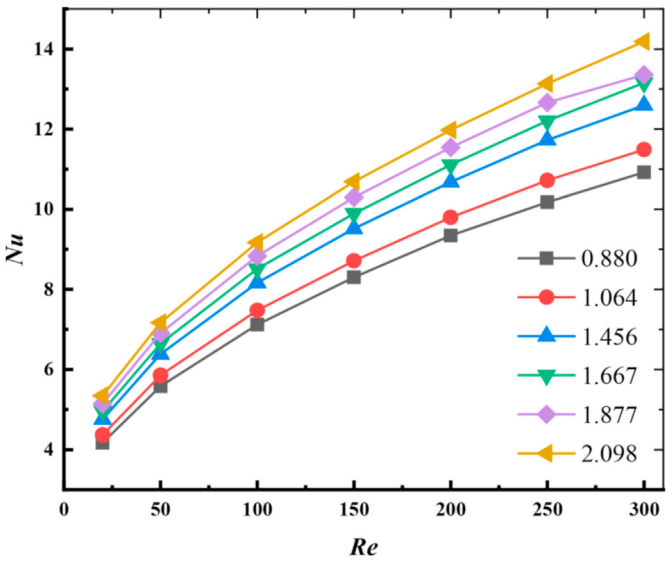


Fig. 19. The effect of a/c on the Nusselt number of oval-shaped capsules. ($20 \leq Re \leq 300$).

capsule has better thermal performance under this geometric parameter.

Fig. 18 shows the effect of a/c on the drag coefficient of the capsule under different Reynolds numbers. It can be seen from Fig. 18 that the trend for the curve is the same as that observed in Fig. 15: the drag coefficient decreases with the increase of Reynolds number under the same geometric parameters, it decreases first and then increases with the increase of a/c under the same Reynolds number. When the Reynolds number is between 150 and 300, the capsule with the ratio of long-radius to symmetry-radius of 1.67 has the minimum drag coefficient.

Fig. 19 shows the effect of a/c on the Nusselt number of capsules when the Reynolds number is between 20 and 300. Unlike the phenomenon in Fig. 16, a/c has a greater influence on the Nusselt number of the oval-shaped capsule. It can be found from the figure that at the same Reynolds number, the Nusselt number increase with a/c, and the difference between the Nusselt number increases with the Reynolds number. But the increasing rate decreases with the increase of Reynolds number. The results also show that the larger the windward area is, the larger the Nusselt number is.

As shown in Fig. 20, the surface-volume ratio and total melting time increase with the increase of a/c, and the average surface heat flux decrease with the increase of a/c. There is a gentle slope in the range of 1.456–1.667. When a/c increases from 0.88 to 2.098, the total melting time increases from 8000 s to 8924 s, and the average surface heat flux decreases from 297.96 W/m^2 to 253.74 W/m^2. The above-mentioned phenomenon indicates the larger the bottom area, the more conducive to the melting of the capsule. This conclusion is consistent with the experimental conclusion in 4.1. Therefore, it is necessary to choose appropriate geometric parameters to balance the flow performance and thermal performance.

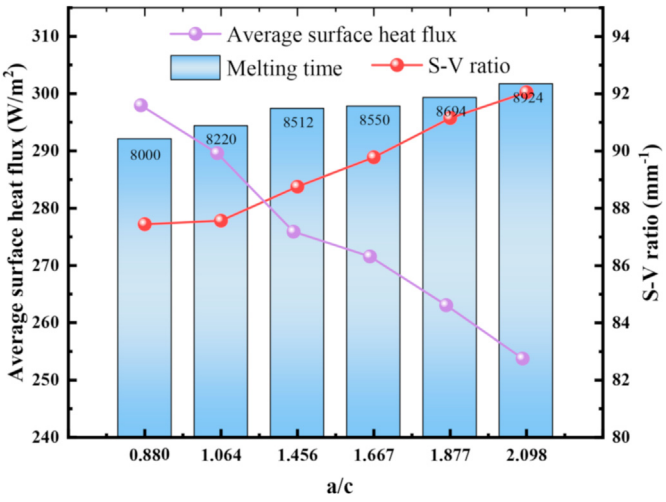


Fig. 20. The effect of a/c on the thermal performance of oval-shaped capsules.

5. Conclusion

To improve the thermal performance and flow characteristics of PCM capsule, the idea of a bio-inspired oval-shaped structure is proposed. The influence of packaging forms (bionic-oval, sphere, and ellipse) and geometric parameters of biomimetic oval PCM capsules on flow and heat transfer process are analyzed by experiment and numerical study. The major conclusions are as follows:

(1) The experimental results show that the constrained and unconstrained melting time of oval-shaped capsule is 4.2% and 12.0% shorter than that of sphere capsule, respectively.
(2) The Nusselt number of the oval structure is 20% higher than that of the sphere structure, and the streamlined structure makes the oval structure have a smaller flow resistance and tail turbulence area.
(3) The oval-shaped capsule has the minimum drag coefficient when a/b and a/c are 1.25 and 1.667, respectively. The Nusselt number increases with the increase of the two proportions, and the latter reveals a greater influence.
(4) The variation of geometric parameters has a certain influence on the thermal performance of the oval structure, and the maximum difference is 10%. It is necessary to choose the appropriate structure according to the actual requirements.

Credit author statement

Dong Yan: Methodology, Writing – original draft. **Wang Fuqiang:** Conceptualization, Writing – review & editing, Funding acquisition. **Zhang Yaqi:** Data curation, Visualization. **Shi Xuhang:** Validation, Formal analysis. **Zhang Aoyu:** Formal analysis. **Shuai Yong:** Investigation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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